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AI-Powered Virtual Mental Health Assistant for Early-Stage NLP-Based Mental Health Screening

Desmond Yong Kiat LOY
School of Computing Science
University of Glasgow
United Kingdom
2837338L@student.gla.ac.uk

Peter ChunYu YAU*
School of Computing Science
University of Glasgow
United Kingdom
PeterCY.Yau@glasgow.ac.uk
*Corresponding author

Dennis WONG
Faculty of Applied Sciences
Macao Polytechnic University
China
cwong@mpu.edu.mo

***Abstract* — Mental health issues have become a significant global concern, affecting individuals' ability to perform daily tasks and increasing the risk of social prejudice. This paper presents a solution to address mental health challenges while reducing the burden on medical professionals. Our research includes a survey of 41 respondents to gauge public perception of mesntal health and their willingness to share their thoughts. We propose a web-based platform that provides mental health guidance and facilitates direct communication with psychologists. The platform integrates a speech-to-text model for audio transcription and a natural language processing (NLP) model to classify mental health conditions. Its architecture ensures secure data storage while enabling users to access essential resources without fear of discrimination.**

Keywords— Natural Language Processing (NLP), RoBERTa, Mental Health, Video Streaming, WebRTC

I. Introduction

Mental health issues have become a major concern in most countries around the world, with many people struggling with anxiety, depression, and other conditions due to various factors such as education and work-related issues. Mental health has become the leading health issue in 31 countries, according to an IPSOS article [1]. Another commentary article from Channel News Asia (CNA) indicates that Singapore's current mental healthcare infrastructure may be unable to handle the increased need for such services with more people willing to act and seek professional help [2]. According to Statista, the issue that adults face worldwide the most is mental health with 44% and cancer being the 2nd largest health problem in 2023 [3]. Therefore, the problem that we are aiming to focus on and tackle is mainly on the following factors, such as challenges in mental health accessibility**,** overburdened mental health professionals and lastly subjective assessments of mental health**.**

Challenges in mental health accessibility could be due to various reasons. According to an NCBI article, it stated many countries around the world that need mental health care lack access to mental support due to a variety of reasons such as manpower crunch [4]. While Singapore, has taken a substantial number of measures to improve mental health challenges, with additional support helplines and even improving the medical system [5, 6] with online support, the issue lies on people with self and social stigma, and potential financial issues may also result in their treatment being delayed. These findings tell that mental health accessibility is influenced due to a variety of complex reasons regardless of whether within Singapore or worldwide.

Overburdened mental health professionals can also be due to factors such as manpower crunch. In Singapore, as of 2022, there were only 203 registered psychiatrists and 212 psychologists in the public sector [7]. As such the use of AI can potentially help to reduce burdens The current measures can still be further improved as technology advances with the inclusion of detecting mental illness with sentiment analysis.

Mental health evaluation often relies on the mental health professional judgment therefore, the assessment may be different between the providers. There was bias due to racism and sexism that led to it being more subjective [8, 9]. Although currently AI chatbots are mainly utilized for providing health guidance and resources, they are not commonly utilized for detecting and tracking potential serious cases that can help to assist mental health professionals.

While mental health remains a critical aspect of overall well-being, existing challenges in accessibility, professional workload, and subjective assessments highlight the need for innovative solutions. With advancements in technology, AI-driven tools can play a crucial role in enhancing mental health support, improving accessibility, and assisting professionals in delivering more efficient and objective care.

II. Literatures

A. Model

While researching for a model to utilize for the model classification purposes. I came across multiple resources that compared the usage of BERT [10], RoBERTa [11], DistilBert [12] and even specialised models for mental health called MentalBERT [13] and MentalRoBERTa[13].

Bidirectional Encoder Representations from Transformers, also known as BERT, is an open-source machine language for natural language processing (NLP) developed by Google in

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2018, with the purpose of handling a widespread number of features and tasks [10].

Robustly Optimized BERT Pretraining Approach also known as RoBERTa is created by Facebook AI research. It was created to optimize the BERT model with the same architecture, allowing for larger dataset management, dynamic masking and improved performance on NLP-related tasks [11]. In a research paper, RoBERTa is shown as the best model with an accuracy of 87.5% which is higher compared to the other BERT-based models like DistilBERT at 86.3% and BERT at 85.3% for the purpose of detecting eating disorders from tweets [14].

DistilBERT is introduced with the purpose of reducing the size of the BERT model while allowing it to be able to perform faster than the BERT model without much loss of capabilities performance [17]. In a research paper that was published, they compared a variety of transformer models for depression on Twitter data, which led to 3 models with relatively high accuracy, on XLNet, DistilBERT and RoBERTa [15].

Although all the reviewed models have demonstrated strong capabilities in NLP tasks, I chose RoBERTa as it seems the most consistent compared to the other models and due to the fact, its accuracy of classification and relatively faster processing compared to BERT in a real-time application like a chatbot will suit better. Although distilBERT also has improvement in speed, they also do come with a slight loss of accuracy compared to RoBERTa as seen in the research paper with the comparison of RoBERTa, DistilBert and another variant of BERT models [14].

B. *Similar Products*

We identified some software created by overseas companies, mainly from in the US. The few software are Clinko[16], SimplePractice[17] and Therapy Notes[18]. I also found a literature paper that propose a software they called "Serene" as a web application-based system to help counteract and detect depression. [19] In Singapore, there is also a specific mental health platform called Mindline.sg [20] that we could utilize to provide guidance and help needed.

Cliniko[16] is a type of management software used by different types of clinicians or health professionals. It contains features like scheduling, treatment records, payment, and performance reports of the system.

SimplePractices[17] is a type of EHR system, designed for behavioural health. It includes features such as scheduling, payment processing, documentation and implementation of Telehealth. The platform also consists of separate Client and Clinician portals that are accessible on the web or mobile.

TherapyNote [18] is an EHR that is also designed for behavioural health. It includes similar features like scheduling, payment and documentation. Therapy notes also include the implementation of telehealth allowing for communication through a secure video appointment.

Serene [19] is a website-based application focused on allowing users to talk and insert questions anonymously without being fearful. It includes guidance videos to help the users to better understand and manage their condition.

Mindline.sg [20] is a Singapore-based site, created as an initiative to better improve the mental health of people. It includes self-help tools that can assist individuals to better improve their mental and emotional well-being. They include an AI chatbot called WYSA which focuses on mental health guidance and support with any possible illness a user might have.

Therefore, based on the research and evaluation done for the models and frameworks, I will be utilizing RoBERTa for my fine-tuning model for the classification of mental illness. The insights gained from the existing platform allow me to better focus on the minimum product features that will help to prioritize user experience, data security and regulatory compliance. This will allow me to provide a more comprehensive solution that can minimally be comparable to the existing product and improve it in a different way.

III. Methodology

A. *Survey Results*

To formulate our methodology to address the mental health issues introduced in Sections I and II, we conducted a survey, with the survey questionnaires shown in Figure 1. The survey receives feedback from 41 students in Singapore.

Fig. 1: Survey Result

The survey feedback of these five questions is shown in Fig. 2 – Fig. 6. It is observed in Fig. 2 that about 63.4% of respondents are satisfied with their mental health.

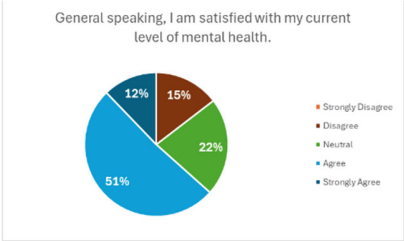

Fig. 2: Response to Survey Question 1 (scale of 1 – 5).

It is observed in Fig. 3 that many respondents (61%) neither agree nor disagree on being worried about their mental health, while 21.9% felt some level of concern during academic studies.

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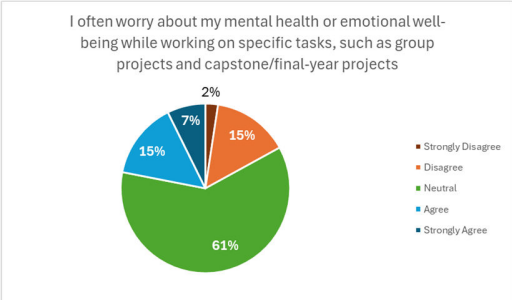

Fig. 3: Response to Survey Question 2

It is observed in Fig. 4 helps to tell that nearly half of the respondents (48.8%) agree on experiencing stress and anxiety when facing academic deadlines.

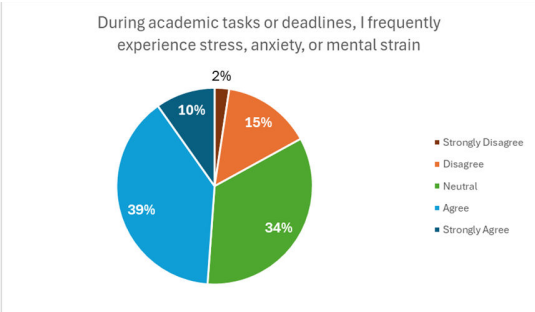

Fig. 4: Response to Survey Question 3

It is observed in Fig. 5, that a significant portion of the respondent (34.1%) disagree and are uncomfortable discussing mental health, while nearly half (48.8%) remained neutral, indicating a possible stigma or reluctance to openly address their mental health issues.

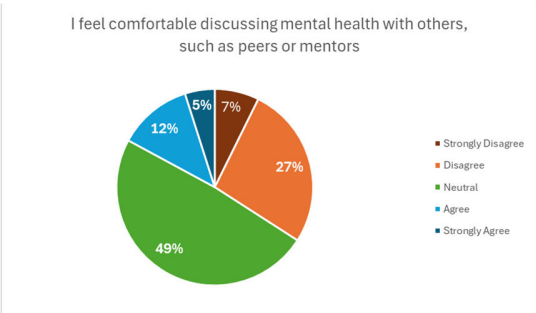

Fig. 5: Response to Survey Question 4 (scale of 1 – 5)

It is observed in Fig. 6, that most of the respondents felt confident in handling mental health challenges while about a third (31.7%) remain uncertain in managing their mental health.

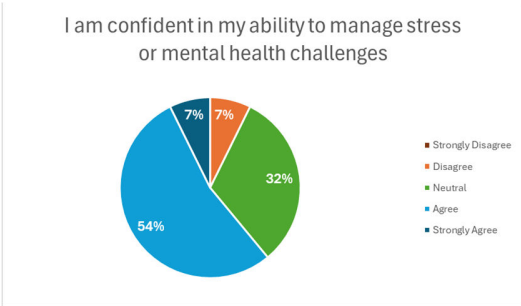

Fig. 6: Response to Survey Question 5

B. *Model training*

For my fine-tuned model, I will be utilizing the RoBERTa base. The dataset I used to fine-tune is from Kaggle called "Sentiment Analysis for Mental Health" [21]. It consists of 53042 data, with 7 classification statuses of anxiety, depression, normal, suicidal, bipolar, stress and per. The dataset is also not balanced with the highest number of data being normal and depression.

Preprocessing of the dataset was performed with a variety of methods like removing missing rows, augmentation, stop words, lemmatization, contractions and even removal of emojis, links and words that are in a tag.

Fine-tuning is done with a pre-classifier layer, with a Gaussian Error Linear unit (GeLU) to help improve the learning dynamics of nonlinearity. Layer normalization is then added to stabilize the process and reduce overfitting with added drop-out to improve generalization before outputting to the 7 logits for the 7 classification statuses.

C. *Mental Health Checklist*

With reference to the 7 classification statuses, I researched their associated mental health assessment checklist to implement as a pre-screening feature for the users. The pre-screening aims to provide users with a self-assessment allowing for early identification of potential mental health concerns.

The chatbot's questionnaire was designed with reference from the following available checklist for mental health:

- PHQ-9 (Patient Health Questionnaire-9) [22] – used for depression screening
- Columbia-Suicide Severity Rating Scale (C-SSRS) [23]– Assess suicidal ideation and risk
- GAD-7 (Generalized Anxiety Disorder-7) [24] – measures anxiety level
- PSS (Perceived Stress Scale) [25] – Evaluate stress
- MDQ (Mood Disorder Questionnaire) [26] – screens for mood disorders, personality disorder

Due to the sensitive nature of some assessments, not all components from these checklists are fully implemented. Instead, a selection of key general questions was used to ensure a non-intrusive yet effective screening experience. The questions that are selected from the checklist are:

1. Have you had trouble sleeping, either too much or too little?

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2. Do you find it hard to concentrate or focus on tasks?
3. Have you been having trouble relaxing recently?
4. Have you noticed changes in your energy levels, whether unusually high or low?
5. Do you feel safe, and do you ever think about hurting yourself or others?

These questions were selected based on relevance to the model and general mental health concerns. The chatbot will then analyse the response based on the replies to the questions using natural language processing to classify user inputs. Fig 8 shows a sample of how the chatbot will show the questions.

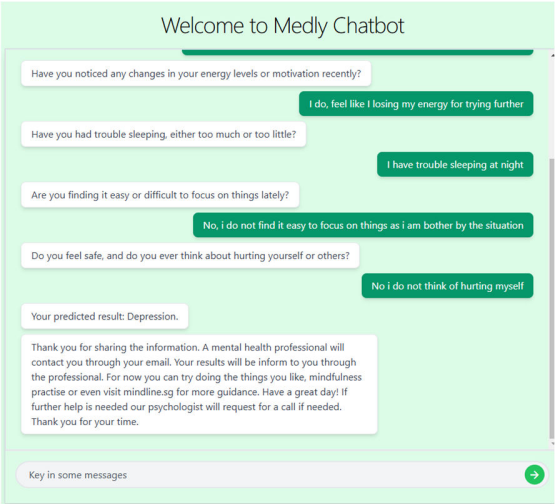

Fig. 8: Chatbot with questions with NLP model classification

After the predicted result is shown, another questionnaire will appear with a scale format. However, only PHQ-9 and GAD-7 are free for public usage after Pfizer release it for free usage hence no permission is required to reproduce, translate, display or distribute the GAD-7 or PHQ-9. [27]

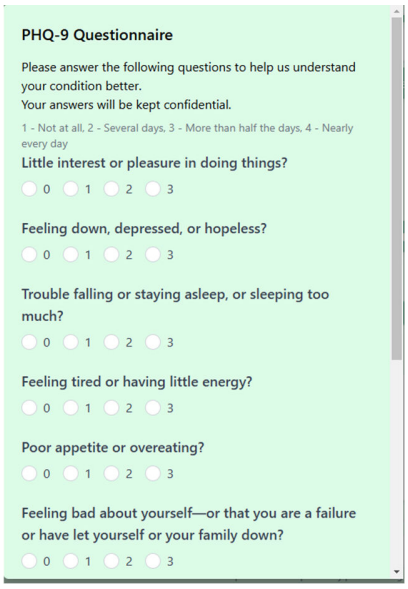

Fig. 9: PHQ-9 Questionnaire form

D. *Proposed design*

For my proposed design., we will be adopting a micro-services architecture with the use of Kubernetes, Django for the front-end, and FastAPI as the back-end services and PostgreSQL as the database. The API services will be running in a distributed microservice architecture. Fig 7 below shows the architecture that I planned to utilize to ensure a robust and secure system.

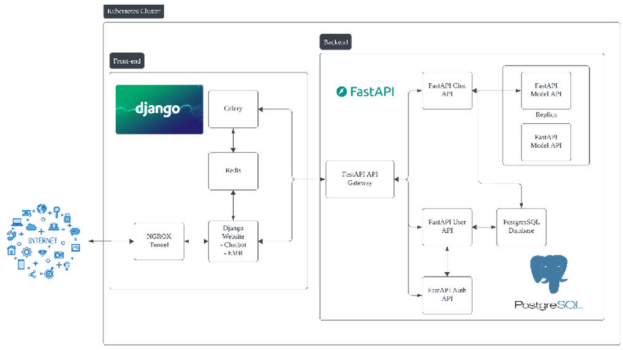

Fig. 7: Architectural Design

In our approach, we are using docker to deploy the container and Kubernetes to manage the services of the application with HorizontalPodAutoScaler. This will help to handle processes that may be overloading and scale the pods accordingly.

For front-end purposes, I will have Django, Celery Worker, and Redis. The Django server will be the front-end server that handles the web pages. Celery is meant to assist with asynchronous task management allowing Django to offload time-consuming operations like chatbot generation and classification. Lastly, Redis is to act as the message broker and cache for the communication between Django and Celery. This allows me to better handle communication synchronously and asynchronously.

For the back-end, all the services are created with FastAPI and the database we utilized is PostgreSQL. Each service serves its purpose, like “model-svc” consists of model-related functions like classification, generating with Gemini API and utilizing the whisper speech-to-text (SST), hence it's configured with HorizontalPodAutoScalar as well.

Based on the methodology, an electronic health record-like system was developed as a self-help system where users can determine whether to get assistance from a psychologist or a chatbot. In the application, the main features will be mainly the chatbot which utilises an NLP model like the image shown in Fig 8. Fig 9 shows the other version of the chatbot where it is utilizing the Gemini API to return a self-help reply.

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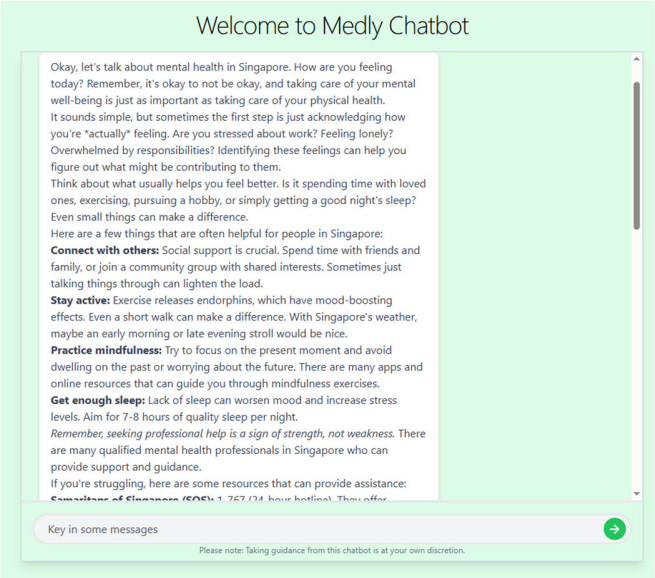

Fig. 10: Chatbot with questions with the Gemini API for guidance

The video consultation includes a transcription of a conversation between the users using a variant of whisper called faster-whisper created by Systran [28] Fig 12 shows an example of how it will look like.

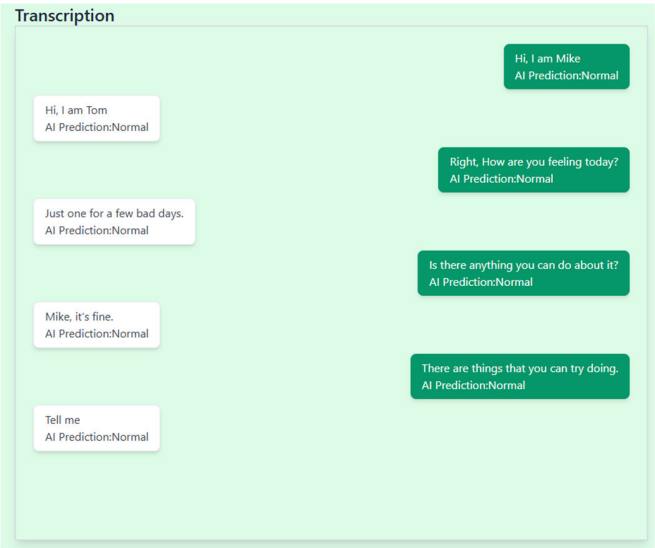

Fig. 12: Video Consultation with WebRTC

Other minor features were also included like a calendar for appointments, user management, audit logs, and role-based access control to manage on the psychologist side.

IV. Result and Analysis

A. Model Training Result and performance Analysis

The model was trained for 5 epochs using the training dataset which comprised 80% of the pre-processed dataset and 10% of the training dataset was used as validation dataset. Table I shows the training and validation performance of each epoch.

TABLE I. Training and Validation Performance per Epoch

Epoch	*Training Loss*	*Training Accuracy (%)*	*Validation Loss*	*Validation Accuracy (%)*
1	1.4034	70.76	1.2041	79.90
2	1.1841	80.54	1.1150	82.13
3	1.0985	84.07	1.0538	85.87
4	1.0436	86.77	0.9948	91.65
5	1.0129	88.69	0.9790	92.79

The model is then tested with the remaining 20% of the dataset obtaining 87.62%, Figure 13 shows the confusion matrix of the result and Table II shows the classification report result.

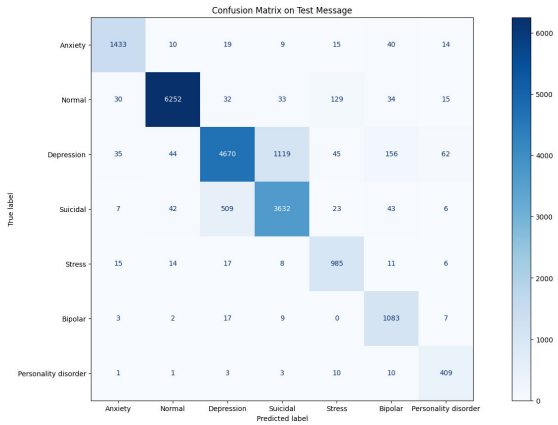

Fig. 13: Confusion Matrix on classification

TABLE II. Training and Validation Performance per Epoch

	Precision	*Recall*	*F1-score*	*Support*
Anxiety	0.94	0.93	0.94	1540
Normal	0.98	0.96	0.97	6525
Depression	0.89	0.76	0.82	6131
Suicidal	0.75	0.85	0.80	4262
Stress	0.82	0.93	0.87	1056
Bipolar	0.79	0.97	0.87	1121
Personality disorder	0.79	0.94	0.83	437

Overall, the result did not show signs of overfitting however Fig 13 shows that there are high rate of misclassification between depression and suicidal, potentially leading to wrong detection.

B. Software Performance Testing Results

To evaluate the performance of the software, a combination of manual testing, unit testing and load testing was conducted. The testing was done across multiple phases, which include:

1) Unit Testing – Focused on API functionalities via pytest and pytest-django

2) Integration Testing & System Testing (Manual) – Conducted through real user interaction to evaluate the end-to-end workflow.

3) Load Testing – Conducted using the Locust library to evaluate system performance under concurrent usage.

Most of the tests are done manually to ensure the functions are integrated correctly with the back-end, especially with the use of WebSocket for the chatbot and video consultation features. With the use of Celery and Redis to handle tasks asynchronously, testing of load directly to API might not

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provide accurate results on the load when running through the website.

V. Conclusion

Through the completion of this website and survey results, we feel that mental health is rather important but taken for granted due to potential stigma and other reasons. Through the creation of this system with AI-driven sentiment analysis and mental health triaging, it can help everyone to better take mental health into their own hands in one way or another, while psychologists could also assist them if needed further help and assistance.

Compared to existing methods, the approach taken here is to focus on assisting others mentally but not forcing a payment requirement on the user and even focus on early detection of potential mental illness with the use of AI technologies. More customization notes systems can potentially help to provide customised treatment plans to help the students in need while providing relevant feedback they can take on to better manage their mental health.

While the minimal system functionality requirement was reached, we still planned to further improve the RoBERTa classification model in terms of accuracy and even add more classification classes in the future to include all possible mental illnesses based on words, additional security features and optimizing resource efficiency overall. Therefore, the overall future work could include:

- Multilingual support
- Real-time emotion analysis to enhance user experience
- Customizable therapy plan for psychologist to customize for each student
- Improve the notification system.
- Increased number of classifications for model to more than 7 mental illness
- Improve user interface and user experience

Overall, this project helps to provide a step towards AI-assisted mental health screening and consultation with further advancements in accessibility and accuracy.

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